



BHABHA UNIVERSITY, BHOPAL

**COURSE CURRICULUM
FACULTY OF SCIENCE**

Postgraduate Programme
(Courses effective from Academic Year 2018-19)

SYLLABUS OF COURSES TO BE OFFERED
“Master of Science”
(Microbiology)

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Dean faculty of Science, Bhabha University

Scheme of Examination

M.Sc. Ist Semester

<u>M.Sc. Ist Semester</u> No. of papers.	<i>Name of papers</i>	<i>E. A.</i>	<i>I. A.</i>	<i>Total Max. Marks.</i>
MB:101	Introduction to Microbiology and General Bacteriology	80	20	100
MB:102	Virology and Mycology	80	20	100
MB:103	Cell Biology and Biochemistry	80	20	100
MB:104	Microbial Genetics and Molecular Biology.	80	20	100
MB:105	Practical	50		50
MB:106	Seminars	50		50
		Total		500

M.Sc. IInd Semester

<u>M.Sc. IInd Semester</u> No. of papers.	<i>Name of papers</i>	<i>E. A.</i>	<i>I. A.</i>	<i>Total Max. Marks.</i>
MB:201	Bioinstrumentation	80	20	100
MB:202	Immunology	80	20	100
MB:203	Microbial Technology	80	20	100
MB:204	Biostatistics , Computer Fundamentals and Bioinformatics.	80	20	100
MB:205	Practical	50		50
MB:206	Seminars	50		50
		Total		500

M.Sc. IIIrd Semester

<u>M.Sc.</u> <u>IIIrd</u> <u>Semester</u> <u>No.of</u> <u>papers.</u>	<i>Name of papers</i>	<i>E. A.</i>	<i>I. A.</i>	<i>Total Max. Marks.</i>
<i>MB:301</i>	<i>Medical Microbiology</i>	<i>80</i>	<i>20</i>	<i>100</i>
<i>MB:302</i>	<i>Recombinant DNA Technology</i>	<i>80</i>	<i>20</i>	<i>100</i>
<i>MB:303</i>	<i>Microbial Physiology and Metabolism</i>	<i>80</i>	<i>20</i>	<i>100</i>
<i>MB:304</i>	<i>Environmental Microbiology</i>	<i>80</i>	<i>20</i>	<i>100</i>
<i>MB:305</i>	<i>practical</i>	<i>50</i>		<i>50</i>
<i>MB:306</i>	<i>Seminars</i>	<i>50</i>		<i>50</i>
		<i>Total</i>		<i>500</i>

M.Sc. IVth Semester

<u>M.Sc. IVth</u> <u>Semester</u> <u>No .of</u> <u>papers.</u>	<i>Name of papers</i>	<i>E. A.</i>	<i>I. A.</i>	<i>Total Max. Marks.</i>
<i>MB:401</i>	<i>Agriculture microbiology</i>	<i>80</i>	<i>20</i>	<i>100</i>
<i>MB:402</i>	<i>Food microbiology</i>	<i>80</i>	<i>20</i>	<i>100</i>
<i>MB:303</i>	<i>Project work of 3-4 months duration</i>			<i>200</i>
		<i>Total</i>		<i>400</i>

MB: 101 INTRODUCTIONS TO MICROBIOLOGY AND GENERAL BACTERIOLOGY

UNIT-I

1. Introduction and history of Microbiology and scope of Microbiology.
2. Microorganism: their general characteristics and composition of microbial world; Prokaryotes and Eukaryotes.
3. Classification of Microorganisms: Haekel's three kingdom concept, Whittaker's five kingdom concept, three domain concept of Carl Woese , classification and salient features of bacteria according to Berger's Manual of Determinative Bacteriology.
4. Nomenclature and modern method of Bacterial taxonomy.

UNIT-II

1. Morphology and ultra structure of bacteria : size, shape, and arrangement of bacteria, ultra structure of bacterial cell wall of eubacteria and archeobacteria , relation of Gram staining to bacterial cell wall. Protoplast and spheroplast formation and L-form.
2. Structure and function of flagella, fimbriae and pilli, capsule- type, composition and function , slime layers, S-layers.
3. Cell membrane- structure and function of bacteria and archeobacteria , mesosomes , ribosomes, nucleoid , cytoplasmic inclusion bodies- polyhydroxy butyrate, polyphosphate granules, oil droplets, cynophycin granules.
4. Endospore: structure, formation and germination of bacterial endospore . Chemotaxis and phototaxis.

UNIT-III

1. Bactrial nutrition: Basic nutritional requirements, growth factors, nutritional categories, physical requirements of bacterial growth.
2. Bacteriological media: types (complex, synthetic, differential and selective media) and their uses, culture characteristics of bacteria on different media.
3. Cultivation of bacteria: aerobic and anaerobic culture, shaker and still culture, maintenance and preservation of microbial culture.
4. Bacterial growth: growth kinetics, growth curve. Batch, continuous and synchronous culture. Measurement of growth and influence of environmental factors affecting growth.

UNIT-IV

1. General concept of Prokaryotic and Eukaryotic genome. E.coli chromosome.
2. Genetic recombination and transformation.
3. Transduction : generalized and specialized transduction, phage conversion.
4. Plasmid: types and their significance. Conjugation and chromosomal mobilization. E.coli as a model prokaryotes.

UNIT-V

1. Control of microorganisms: Microbial death curve, concept of bioburden, thermal death time and decimal reduction time. Factors influencing the effectiveness of antimicrobial agents.
2. Control of microorganisms by physical agents: heat , filtration and radiation.
3. Chemical control of microorganisms : Halogens, phenol and other phenolic compounds, heavy metals, alcohols, ethylene oxide and aldehydes.
4. Antibiotics: properties and mode of action, drug resistance and its significance. Antimicrobial sensitivity test.

MB: 102 VIROLOGY AND MYCOLOGY

UNIT-I

1. Brief outline on discovery and origin of viruses.
2. General properties of viruses, morphology and ultra structure of viruses, capsid and their arrangements, types of envelopes and their composition, measurement of viruses.
3. Viral genome; their types and structure, viral related agents-viroids and prions.
4. Classification and general properties of major families of viruses including detail account of their mode of replication.

UNIT-II

1. Cultivation of viruses- in embryonated eggs, experimental animals and cell lines; primary and secondary cell lines, diploid cell culture.
2. Assay of viruses: physical and chemical methods (protein , nucleic acid, radioactivity tracers, electron microscopy, plaque method, pock counting and end point method.)
3. Serological methods: hemagglutination, hemagglutination inhibition, neutralization test, complement fixation, ELISA, RIA and immunofluorescence assay (IFA).
4. Purification of viruses: gradient centrifuge, electrophoresis, and chromatography.

UNIT-III

1. Plant viruses: recent advance in classification of plant viruses. Structure and pathogenicity of TMV.
2. Transmission of plant viruses with vector (insect, nematodes and fungi) and without vector (contact, seed and pollens). Biochemical changes induced by virus in plant cell.
3. Animal viruses: nomenclature and classification of animal viruses. Host responses to viral infection.
4. General idea about Cyanophage, Actinophage, and Mycophage.

UNIT-IV

1. Bacteriophage: classification, morphology and ultra structure.
2. One step growth curve (latent period, eclipse period, and burst of size.)
3. Life cycle : lytic and lysogenic life cycle of bacteriophages.
4. Brief account of M13, Mu, T3, T4, φx174 and λ phage.

UNIT-V

1. Structure, reproduction and classification of fungi, general characteristics of Zygomycetes, Ascomycetes, Basidiomycetes, and Deuteromycetes.
2. Cultivation of fungi, culture media for fungal growth, effects of environment on growth, isolation, identification and preservation of fungi.
3. Dimorphic fungi, yeast morphology, general characteristics and reproduction. Lichens, Micorrhiza, and Actinomycetes.
4. Ecology of fungi: concept of fungistasis, fungicidal, antagonism, symbiosis and Synergism

MB: 103- CELL BIOLOGY AND BIOCHEMISTRY

UNIT-I

1. Cell: size, shape, types & chemical composition of the cell.
2. Structural organization and function of intracellular organelles of eukaryotic cell: nucleus, mitochondria, golgi body, lysosomes, endoplasmic reticulum, peroxisomes, plastids, chloroplast, vacuole, cytoskeleton.
3. Membrane structure and function: molecular organization of cell membrane, membrane models, mechanisms of intracellular transport.
4. Cellular interaction: differentiation of cell membrane and intracellular communication and Gap junction.

UNIT-II

1. Cell differentiation: general characteristics of cell differentiation and cytoplasmic factors, differential gene action.
2. Cell signaling: cell surface receptors, G-protein, signal transduction pathways.
3. Cell cycle: mitosis and meiosis and their regulation. Programmed cell death and apoptosis.
4. Cancer biology: characteristics of cancer cell, types of cancer, oncogene and Tumor markers.

UNIT-III

1. Carbohydrates: structure of sugars, classification, properties, chemical reactions, stereoisomerism and optical isomers of sugars.
2. Structure, properties and function of disaccharides, oligosaccharides, and polysaccharides, carbohydrate derivatives; peptidoglycan, glycoproteins, glycolipids.
3. Lipids : classification, structure, properties and functions of fatty acids, triacylglycerols, phospholipids, sterols and terpenes.
4. Lipids with specific biological functions, micelles and liposomes.

UNIT-IV

1. Amino acids: structure, classification, properties and functions.
2. Proteins : structural and functional proteins, synthesis of peptide bonds. Primary, secondary, tertiary and quaternary structure of proteins.
3. Nucleic acids: structure and properties of purines and pyrimidine bases, nucleosides and nucleotides.
4. Basic structure and types of DNA and RNA.

UNIT-V

1. Enzymes: basic concept as a biocatalyst, specificity, active sites, activity unit and isoenzymes, enzyme classification.
2. Enzyme kinetics- Michaelis-Menton equation for simple enzymes, determination of kinetic parameters.
3. Enzyme inhibition: competitive, noncompetitive and uncompetitive inhibition, allosteric enzymes.
4. Vitamins and cofactors: structure, distribution and biological properties.

MB:104 MICROBIAL GENETICS AND MOLECULAR BIOLOGY

UNIT-I

1. Organization of genetic material in prokaryotes and eukaryotes.
2. Concept of gene, genome, genome size, C-value, and C-value paradox.
3. Nucleic acid as a genetic information carriers; experimental evidence. DNA denaturation and renaturation.
4. Gene is a unit of mutation and recombination; molecular basis of mutations, physical and chemical mutagens, spontaneous and induced mutation, selection of mutant.

UNIT-II

1. DNA damage and repair: types of DNA damage (deamination, oxidative damage, alkylation and pyrimidine dimers.), repair mechanism; mismatch repair, nucleotide excision repair, recombination repair, SOS repair.
2. Structure of DNA, super helicity of DNA, linking number, topological properties and role of topoisomerase.
3. DNA replication: general principle, various mode of replication, unwinding of DNA Helix, continuous and discontinuous synthesis of leading and lagging strands.
4. Enzymes of DNA replication in prokaryotes and eukaryotes; DNA polymerases, DNA ligase, primase.

UNIT-III

1. Structural features of RNA (rRNA, tRNA, mRNA) and polycistronic and monocistronic RNA.
2. Transcription: general principle and processes of transcription; initiation, elongation and termination, types of RNA polymerases, inhibitors of RNA synthesis.
3. Control of Transcription by interaction between RNA polymerases and promoter region, use of alternate sigma factors, controlled termination; attenuation and antitermination.
4. Post transcriptional modification, maturation and splicing of RNA transcripts, catalytic RNA.

UNIT-IV

1. Genetic code: nature of genetic code, codon, anticodon, wobble hypothesis.
2. Protein synthesis: steps, details of initiation, elongation and termination.
3. Inhibitors of protein synthesis: signal hypothesis.
4. Post translational modification: covalent modification, phosphorylation, glycosylation, methylation. Protein targeting.

UNIT-V

1. Regulation of gene expression: operon concept; regulatory and structural gene, operator, promoter, repressor, induction and repression, positive and negative control.
2. Lac-operon, ara-BAD operon, trp operon, attenuation, mechanism of regulation of transcription.
3. Regulation of gene expression in eukaryotes: Britton and Davidson's model of regulation involve HCP and NHCP and hormones.
4. Transposable elements.

I Semester

Practical List

- 1) General Instrumentation
- 2) Preparation of Media
- 3) Staining Techniques: Differential and Structural Staining Methods
- 4) Isolation Techniques
- 5) Determination of Standard Plant Count.
- 6) Operation of *Mcparland* Scale
- 7) Study of Bacterial Growth-
 - A) Factors Affecting Bacterial Growth.
 - B) Growth Curve of Bacteria, Calculation Of Growth Rate And Generation Time (Demonstration).
 - C) Diauxic Growth Pattern of Bacteria (Demonstration)
- 8) Anaerobic Cultivation-
 - A) Anaerobic Methods-Gaspak Anaerobic Jar/Candle Jar Method.
 - B) Isolation of Clostridium from Soil and Its Identification.
- 9) Microbiological Study of Actinomycete-
 - A) Isolation and Characterization of Actinomycetes
 - B) Cover slip Culture Study for Study Of Morphological Structure Of Antinomycetes.

MB : 201 BOINSTUMENTATION

UNIT-I

1. *Microscopy: history and principles of microscopy, magnification power, Resolution limit, resolving power, numerical aperture.*
2. *Principles and application of light microscopy, properties of light, bright field, dark field, phase contrast and fluorescent microscopy. Determination of size of microorganisms by micrometry.*
3. *Principles and application of electron microscopy- transmission and scanning electron microscopy.*
4. *Newer techniques in microscopy- confocal microscopy, scanning probe microscopy (scanning tunneling microscope and atomic force microscope).*

UNIT-II

1. *Concepts of acids and bases, pH, measurement of pH by uses of indicator and electronic meter, buffersystem.*
2. *Chromatography: principles, types and applications of partition, adsorption, gel filtration, paper and thin layer chromatography.*
3. *Affinity, ion exchange, and gas chromatography.*
4. *High performance liquid chromatography and FPLC.*

UNIT-III

1. *Electrophoresis: principle, types and applications, frontal and zonal electrophoresis, paper, starch gel, Polyacrylamide and agarose gel electrophoresis.*
2. *Isoelectric focussing and Isotachophoresis.*
3. *Two dimentional gel electrophoresis and pulse field gel electrophoresis.*
4. *Immunological techniques: immunoelectrophoresis , immunodiffusion, immuno fluorescence.*

UNIT-IV

1. *Spectroscopy: basic principles, law of absorption and radiation, principles and application of visible,ultraviolet, infrared and mass spectroscopy.*
2. *Principles and application of NMR and ESR.*
3. *Principles and application of colorimetry, fluorscence flame photometry.*
4. *Fluorimetry , polarimetry and turbidometry.*

UNIT-V

1. *Centrifugation: basic principles of analytical and preparative centrifuge, differential and density gradient, zonal and isopycnic centrifuge. Sedimentation coefficient, factors affecting sedimentation coefficient and application.*
2. *Radioisotope techniques: half life, radioactive decay, radioactive assay methods based on ionization and excitation of gases.*
3. *Geiger Muller counter, liquid scintillation counter and gamma counter. Quenching and use of radioisotopes in biological systems. Autoradiography- principles and applications.*

MB: 202 IMMUNOLOGY

UNIT-I

1. *Infection: types of infection, sources of infection, reservoirs and vehicles of infection, predisposing factors.*
2. *History of immunology, development of immunology as disciplines.*
3. *Immune response: mechanism of innate and adaptive immune response.*
4. *Hematopoiesis: development of immune cells, regulation of hematopoiesis and stem cell technology.*

UNIT-II

1. *Structure, composition and types of cells involved in immune response: mononuclear cells, granulocytes, antigen presenting cells, lymphoid cells.*
2. *Anatomical organization of immune system: primary and secondary lymphoid organs: structure and function.*
3. *Antigens- structure and properties, factors affecting the immunogenicity, properties of B and T- cell epitopes, haptens, mitogens, superantigen.*
4. *Antibody: structure, properties, types and function of antibodies, antigenic determinants on immunoglobulin; isotypes, allotypes, and idiotypes.*

UNIT-III

1. *Hybridoma technology and monoclonal antibody production and characterization.*
2. *Application of monoclonal antibodies in diagnosis, therapy and basic research, antibody engineering.*
3. *Antigen- antibody interaction : avidity and affinity measurements, detection of antigen-antibody interaction by precipitation, agglutination, RIA, ELISA, immunodiffusion.*
4. *Major histocompatibility complex: organization of MHC genes, types and function of MHC molecules, MHC polymorphism, MHC related diseases.*

UNIT-IV

1. *Complement system: components, activation pathways, regulation of activation pathways, role of complement system in immune response.*
2. *Cytokines: types, structure and functions, cytokine receptors, cytokine regulation of immune receptors.*
3. *Immune response to infectious diseases: viral infection, bacterial infection, protozoan diseases, helminthes related diseases.*
4. *Vaccines: Active and passive immunization, whole organ vaccine, purified macromolecule as a vaccine, DNA vaccine, recombinant vaccine.*

UNIT-V

1. *Hypersensitivity: type I, II, III and type IV hypersensitivity. Immunodeficiency diseases: primary and secondary immunodeficiency.*
2. *Autoimmunity: organ specific autoimmune diseases, mechanism of autoimmune diseases and therapeutic approaches.*
3. *Transplantation immunology: immunologic basis of graft rejection, clinical manifestation of graft rejection and clinical transplantation.*
4. *Cancer immunology: tumor antigen, immune response to tumor, oncogene and induction, cancer immunotherapy.*

MB: 203 MICROBIAL TECHNOLOGY

UNIT-I

1. *Industrial important strains of bacteria, fungi, and actinomycetes .Novel microbes for future industry.*
2. *Isolation and screening of the industrially important strain from diverse ecosystem.*
3. *Method of strain improvement, mutagenesis, strain breeding by protoplast fusion, sexual and para sexual recombination.*
4. *Fermentation technology: principles of fermentation. Fermenter and bioreactors: monitoring and control of parameters, designing, operation and application.*

UNIT-II

1. *Downstream processing: filtration of fermentation broths recovery of biological products by distillation, superficial fluid extraction.*
2. *Detection, analysis and quality control of fermentation products and raw materials.*
3. *Industrial production of alcohols: vinegar, wine and alcohol.*
4. *Industrial production of solvents-glycerol, acetone, and Butanol.*

UNIT-III

1. *Industrial production of citric acid and glutamic acid.*
2. *Microbial production of enzyme of industrial important: amylase and proteases.*
3. *Methods of whole cell immobilization, enzyme immobilization and application.*
4. *Industrial production of antibiotics, penicillin and streptomycin.*

UNIT-IV

1. *Hygiene and safety in fermentation industries.*
2. *Microbial production of Vitamin B and B₁₂.*
3. *Microbial production of Interferon, Insulin, flavours and fragrances.*
4. *Bioelectronics : Biochips and biosensors.*

UNIT-V

1. *Microbial production of vaccines.*
2. *Microbial production of polymers : Dextran and xanthan.*
3. *Microbial transformations : Steroid biotransformation*
4. *Intellectual property rights (IPR) and protection (IPP)*

MB: 204 Biostatistics , Computer Fundamentals and Bioinformatics

Unit I

1. *Introduction to Biostatistics, Common terms, notions and Applications*
2. *Statistical population and Sampling Methods*
3. *Classification and tabulation of Data*
4. *Diagrammatic and graphical presentation*
5. *Frequency Distribution, Measures of central value*
6. *Measures of variability; Standard deviation, standard Error, Range, Mean Deviation, Coefficient of variation, Analysis of variance*

Unit II

1. *Basic tests, Test of significance; t-test, chi-square test.*
2. *Regression; Basic of regression, regression analysis, Estimation, Testing, prediction, checking and residual analysis.*
3. *Multivariate Analysis.*
4. *Design of Experiments, randomization, replication, local control, complimentary
Randomized, randomized block design*

Unit III

1. *Factor Analysis.*
2. *Path analysis*
3. *Introduction to data mining*
4. *Virtuous Cycle.*

Unit IV

1. *Classification and Discriminant Analysis Tools: CART, Random forests,*
2. *Fisher's discriminant functions.*
3. *Neural networks.*
4. *Multilayer perception, predictive ANN model building using back propagation algorithm, exploratory data analysis.*

Unit V

1. *Introduction to computer basics, concept of hardware windows XP and LINUX*
2. *Concept of file, folders, directories and their management by windows XP and LINUX*
3. *Office applications : MS- Office, MS- Word, MS- Excel and MS- PowerPoint*
4. *Open Office on Linux: Word Processor, spread sheets, Impress*
5. *Statistical Packages: Sigma plot etc.*
6. *Introduction to bioinformatics*
7. *Internet- introduction and application*
8. *Statistical analysis software*

II SEMESTER
PRACTICAL LIST

- 1) Isolation of Genomic DNA from Bacterial Cell And Its Analysis.
- 2) Isolation of RNA from Yeast Cell And Its Analysis.
- 3) Study of UV Absorption Spectra of Nuclei Acid.
- 4) To Check Purity of DNA by Spectrophotometric Method.
- 5) Study of Lethal Action of Ultraviolet Radiation.
- 6) Solution of Mutants by Replica Plant Method.
- 7) Isolation of Drug Resistant Mutants By Gradient Plant Method
- 8) Study of Transfer of Antibiotic Resistance Between Bacterial Species By Conjugation Process.
- 9) Isolation of Photosynthetic Bacteria.
- 10) Chromatography
 - A) Paper Chromatography
 - B) Thin Layer Chromatography
 - C) Gel Filtration Chromatography
 - D) Ion Exchange Chromatography
- 12) Separation of Cell/Cell Organelles By Density Gradient Centrifugation
- 13) Electrophoresis
 - A) Agarose Gel Electrophoresis
 - B)SDS-PAGE

MB : 301 MEDICAL MICROBIOLOGY

UNIT-I

1. *Early discovery of pathogenic microorganisms, development of bacteriology as scientific discipline, contribution made by eminent scientists.*
2. *Host-parasite relationship governing the infection and establishment of disease, factors affecting virulence.*
3. *Normal microflora of human body; normal flora of skin, respiratory, gastrointestinal, genital tract, role of resident flora*
4. *Mode of spread of infection; Respiratory, skin, wound & burn infection, venereal infections, alimentary tract infection, blood born infection.*

UNIT-II

1. *Source of infection, Pathogenicity, Epidemiology & Lab diagnosis of *Pneumococcus*, *Neisseria*, *Corynebacterium*, & *Clostridium*.*
2. *Source of infection, Pathogenicity, & Epidemiology & Lab diagnosis of members of the family Enterobacteriaceae, Coliform; Proteus, Shigella, Salmonella .*
3. **Vibrio*, *Mycobacterium*, *Staphylococcus*, *Pseudomonas*, *Pasteurella*, *Yersenia*, *Brucella*.*
4. **Actinomycetes*, *Rickettsiaceae* , *Chlamydiae*, *Spirochaetes*.*

UNIT-III

1. *Morphology , pathogenesis, immune response, diagnosis and prevention of Pox viruses , Herpes Simplex type I and type II, Picorna viruses.*
2. *Paramyxoviruses, Measles & Mumps viruses & Rhabdoviruses.*
3. *Hepatitis viruses, Arboviruses, Orthomyxoviruses.*
4. *Oncogenic viruses & HIV virus & Prions.*

UNIT-IV

Pathogenesis, Life Cycles, Immunity, Disease produced, Diagnosis & Prophylaxis of

1. **Plasmodium vivax*, *falciparum*, malariae.*
2. **Entamoeba histolytica* & *Coli**
3. **Toxoplasmosis* , *Trypanosomiasis* & *Leishmaniasis*.*
4. **Roundworm* & *Tapeworm*; *Taenia solium* & *segineta*.*

UNIT-V

1. *Fungal infection: description & classification of pathogenic fungi*
2. *Infection caused by dermatophytes (*Microsporum*, *Trichophyton* & *Epidermatophyton*)*
3. *Definition, Causative agent, Source of infection, Epidemiology, Symptomatology & Diagnosis of Aspergillosis & Candidiasis*
4. *Source of infection, Epidemiology, Symptomatology, & Diagnosis of Blastomycosis, Histoplasmosis.*

UNIT-I

1. *Enzymes used in DNA technology: Restriction and modification enzymes, nucleases, polymerases, ligase, kinases and phosphatases .*
2. *Cloning vectors: Plasmids , Phagmids, Cosmids, Artificial chromosomes, Shuttle vectors , Expression vectors*
3. *Cloning Techniques: Isolation & purification of genomic & plasmid DNA & RNA ,Gel electrophoresis of nucleic acids (RNA & DNA); Pulse field gel electrophoresis.*
4. *Construction of genomic and cDNA libraries.*

UNIT-II

1. *Screening of clones from libraries :Expression based screening , Interaction based screening ,Preparation of probes , Restriction mapping.*
2. *Principles of hybridizations and hybridization based techniques: Colony, plaque, Southern, Northern and in situ hybridizations.*
3. *Western and southwestern blotting ,Microarray based detections and RNA interferon.*
4. *Characterization of clones : DNA sequencing , S1 nuclease and RNase mapping.*

UNIT-III

1. *Oligonucleotide synthesis.*
2. *Principles & applications of Polymerase Chain Reaction (Types).*
3. *DNA fingerprinting.*
4. *Mutagenesis : Site directed mutagenesis, Transposon mutagenesis.*

Unit IV

1. *Gene transfer techniques: Electroporation and microinjection, Transfection of cells: Principles and methods.*
2. *Germ line transformation in Drosophila and transgenic mice: Strategies and methods.*
3. *Construction of knockout mutants.*
4. *Promoter characterization: promoter analysis through reporter genes, electrophoretic mobility, shift assay, DNA foot-printing.*

UNIT-V

1. *Applications of Recombinant DNA Technology : Monitoring of gene expression in live cells, Crop and livestock improvement .*
2. *Molecular diagnostics, Biosafety & ethical considerations.*
3. *Gene therapy: somatic and germ line gene therapy; DNA drugs and vaccines.*
4. *Transgenic technologies and there use in microbial technology.*

MB: 303 MICROBIAL PHYSIOLOGY AND METABOLISM

UNIT-I

1. Basic concept of bioenergetics and metabolism.
2. First and second law of thermodynamics, concept of free energy, entropy and enthalpy.
3. High energy phosphate compounds, role of ATP, ATP cycle, structural basis of free energy change during hydrolysis of ATP.
4. Biological oxidation and reduction reaction, role of reducing power.

UNIT-II

1. Carbohydrate metabolism: glycolysis and its regulation, Feeder pathway of glycolysis and carbohydrate –homo and heterolactic fermentation. Glycogenesis, Glycogenolysis. Gluconeogenesis ; pathways and regulation.
2. Pentose phosphate pathway, kreb's cycle and glyoxalate pathway.
3. Substrate level phosphorylation and oxidative phosphorylation, electron transfer reaction in mitochondria, electron carriers and multienzyme complex I to IV.
4. ATP synthesis: chemiosmotic theory, shuttle system, regulation of oxidative phosphorylation and uncouplers, inhibitors of oxidative phosphorylation.

UNIT-III

1. Photosynthesis: structure of chloroplast, light reaction and dark reaction; Calvin cycle, C3 and C4 pathway.
2. Mechanism of energy generation in cyanobacteria, green bacteria and purple sulphur bacteria and chemolithotrophs.
3. Lipid metabolism digestion absorption; oxidation of unsaturated fatty acid and odd chain fatty acid, ketone bodies.
4. Lipid biosynthesis: biosynthesis of fatty acids, triacylglycerol and phospholipids and regulation of fatty acid metabolism.

UNIT-IV

1. Amino acid metabolism: biosynthetic families of amino acids.
2. Breakdown of amino acids into six common intermediates and urea cycle and regulation of amino acid metabolism.
3. Nucleotide metabolism; biosynthesis of purines and pyrimidines nucleotide by de novo and salvage pathways.
4. Degradation of purines and pyrimidines nucleotides.

UNIT-V

1. Nitrification, denitrification and pathways of nitrate and ammonia assimilation. Nitrogen cycle.
2. Assimilation of nitrogen: denitrogen fixation- free living and symbiotic, diazotrophic organisms.
3. Biochemistry of nitrogen fixation: nitrogenase complex, function of nitrogenase, regulation of nitrogenase by oxygen and combined nitrogen sources.
4. Genetics of nitrogen fixation; nif genes and their regulation.

MB: 304 ENVIRONMENTAL MICROBIOLOGY.

UNIT-I

1. *Microbial ecology: basic concepts, types and microbial habitats, factors affecting microbial population.*
2. *Microbial interactions: competition, amensalism, parasitism, mutualism, commensalisms, synergism.*
3. *Biogeochemical cycles: carbon, nitrogen, phosphorus and sulphur cycle.*
4. *Conservation and management of microbial diversity: biodeterioration and biodegradation.*

UNIT-II

1. *Microbiology of air: microorganism of air, enumeration of air micro flora.*
2. *Significance of air micro flora.*
3. *Brief account of air borne transmission of bacteria, fungi , pollens and viruses.*
4. *Air borne diseases and their prevention.*

UNIT-III

1. *Soil microbiology: microflora of soil: soil microorganisms associated with plants: rhizosphere, mycorrhizae.*
2. *Role of microorganisms in organic matter decomposition (cellulose, hemicellulose, lignin).*
3. *Bioleaching; introduction, application of bacterial leaching leaching techniques, properties of bioleaching.*
4. *Microbial degradation of xenobiotics , petroleum and oil spills in environmental decay behaviours and degradative plasmid.*

UNIT-IV

1. *Water microbiology: aquatic microorganisms; fresh water and sea water microflora. Microorganisms and water quality, water pollution.*
2. *Water purity test and indicator organisms, method used in environmental studies –BOD, COD, DO.*
3. *Common water born disease and their control measure.*
4. *Water purification: flocculation, chlorination and purification.*

UNIT-V

1. *Microbiology of waste water and effluent treatments, aerobic process : primary , secondary and tertiary treatment : trickle filter ,oxidation ponds and stabilization ponds , principle of aerobic digestion.*
2. *Bioremediation of contaminations.*
3. *Extremophiles –acidophilic, alkalophilic, thermophilic microbes with adaptation and application in ecosystem.*
4. *Microbial biofilms : physiology ,morphology, biochemisty of microbial biofilms, mechanism of microbial Adherence , beneficial and harmful role of biofilms.*

III SEMESTER
PRACTICAL LIST

- 1) Restriction digestion of DNA samples using restriction endonucleases.
- 2) DNA fingerprinting technique-restriction Fragment length polymorphism-RFLP
- 3) To perform ligation of DNA fragments
- 4) DNA purification from electrophoresed agarose gel (electro elution/silica-based technique, etc)
- 5) DNA amplification by polymerase chain reaction (demonstration)
- 6) DNA fingerprinting techniques-random amplified polymorphic DNA-RAPD.
- 7) Blotting techniques (demonstration)
- 8) Demonstration of antibiotic susceptibility pattern of pathogenic microbes.
- 9) Study Of Synergistic Additive Effect Of Antibiotics.
- 10) Isolation and Identification of Resident Normal Flora From Skin/Throat.
- 11) Effect of Disinfectant On Microflora Of Skin
- 12) Haematology: RBC Count, Total WBC Count, Differential WBC Count, Hemoglobin Estimation.
- 13) Laboratory Analysis of Urine-Physical, Chemical, Microscopic and Bacteriological Analysis.
- 16) Isolation, Biochemical Characterization and Identification Of Medically Important Bacteria-
*E.Coli, Klebsiella Pneumonia, Salmonella Typhi, Proteus Mirabilis, Pseudomonas aeruginosa.
- 17) Microbial Analysis of Air
- 18) Quantitative Analysis Of Microorganisms Present In Soil
- 19) Evaluating the Soil Health of Agricultural Soil (Demonstration)
*pH, Organic Carbon, Phosphorus Potassium, Ammoniacal- Nitrogen, Nitrate Nitrogen.
- 20) Bacteriological Analysis of Water: Qualitative And Quantitative Methods-
A) Standard Plate Count (SPC)
B) Most Probable Number (MPN) Of Coliform.

C) Routine Coliform Test-Presumptive, Confirmed, Completed Test

D) Membrane Filtration Technique

21) Sewage Analysis-

A) Bacteriological Analysis Of Sewage-Qualitative And Quantitative Methods

B) Determination Of Indicators Of Pollution By Measuring BOD/COD Of Different Effluent.

MB: 401 AGRICULTURAL MICROBIOLOGY

UNIT I

1. *Microorganisms of soil*
2. *Rhizosphere and phyllosphere microflora*
3. *Brief account of Microbial interactions: antagonism, symbiosis, mutualism, commensalisms, synergism and parasitism.*
4. *Nutrient cycle : Carbon cycle , nitrogen cycle, phosphorous cycle and sulphur cycle.*

UNIT II

1. *Role of enzymes and toxins in pathogenesis.*
2. *Fungal diseases of plants: Rusts of wheat, linseeds; late blight of potato; red rot of sugarcane.*
3. *Bacterial diseases of plants : Citrus canker, blight of rice*
4. *Viral diseases of plants : Leaf curl of Papaya, vein clearing of lady's finger*

UNIT III

1. *Physical and chemical control of plant diseases.*
2. *Bacterial control of insect pests : Bacillus thuringiensis as bacterial insecticide*
3. *Viral control of insect pests : Nuclear polyhedrosis viruses (NPV) and cytoplasmic polyhedrosis virus (CPV), Metarhizium anisopliae.*
4. *Fungal control of insect pests : Entomopathogenic fungi : Beauveria bassiana, Verticillium lecani, Hirsutella thompsoni*

UNIT IV

1. *Storage fungi: Categories of storage fungi, conditions during storage in relation to damage of seeds, harmful effects*
2. *Mycotoxins and their effect on human being.*
3. *General idea about quarantine*
4. *Production of biogas and alcohol from agricultural wastes*

UNIT V

1. *Biofertilizers : Types, production and application*
2. *Mycorrhizae : Types and their application in agriculture and forestry.*
3. *Vermicomposting*
4. *Reclamation of waste agricultural land by microorganisms.*

MB: 402 FOOD MICROBIOLOGY

UNIT I

1. *Microorganisms important in food microbiology: molds, yeast and bacteria –general characteristics, classification and importance.*
2. *Principles of food preservation, preservation by use of high temperature, low temperature, drying and dessication.*
3. *Chemical preservatives and additives.*
4. *Preservation by radiation.*

UNIT II

1. *Factors influencing microbial growth in food: Extrinsic and intrinsic factors.*
2. *Microbial spoilage of food. Chemical changes caused by the microorganisms during spoilage.*
3. *Spoilage of fish, meat, poultry, eggs, fruits and vegetables.*
4. *Detection of spoilage and characterization.*

UNIT III

1. *Classification of food borne diseases.*
2. *Food borne infections : Brucella, Bacillus, Clostridium perfringens, vibrio, yersinia Escherichia, Salmonella, Shigella.*
3. *Food intoxication: Staphylococcal intoxication, Clostridial poisoning.*
4. *Food adulteration and prevailing food standards in India .*

UNIT IV

1. *Microbiology of Milk : Sources of microorganisms in milk and types of microorganisms in milk.*
2. *Microbiological examination of milk (standard plate count, direct microscopic count, reductase, and phosphatase test).*
3. *Dehydration and pasteurization of milk.*
4. *Dairy products from microorganisms : Butter , yoghurt and cheese .*

UNIT V

1. *Microorganisms as source of food : Single Cell Protein (SCP)*
2. *Mushrooms and food value of mushrooms*
3. *Food conversions : Lactic acid conversions, soyabean conversions and Bakery*
4. *Microbiological estimation of food : Sample collection.*